What is a Senior Software Engineer in Big Tech?

How the role is typically defined by Big Tech and at larger tech companies, and frameworks for thinking about seniority.

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What is a senior engineer, and what are typical expectations at the senior level?

This is a question several readers have been asking following the issue, [What is a Staff+ engineer?](#)

In my observation, a senior engineer is basically someone who gets things done without much guidance. But it's more nuanced. "Getting things done without much guidance" means different things at different types of companies:

- At a digital agency where the 'what' is pretty well defined by the client
- At a small company where everyone knows everyone else, and people rarely get blocked by others
- At a 10,000-person tech company where "getting things done" tends to involve working with several teams, and it's common enough to get blocked by other teams

We'll examine this topic by breaking down what "senior engineer" means at different places: starting with Big Tech and large tech companies which employ well over 1,000 software engineers. I gathered details for the definitions by talking with current engineers at such companies, reading expectations outlined within the company, or – in the case of Dropbox – relying on a publicly shared engineering career framework.

In today’s issue, we cover:

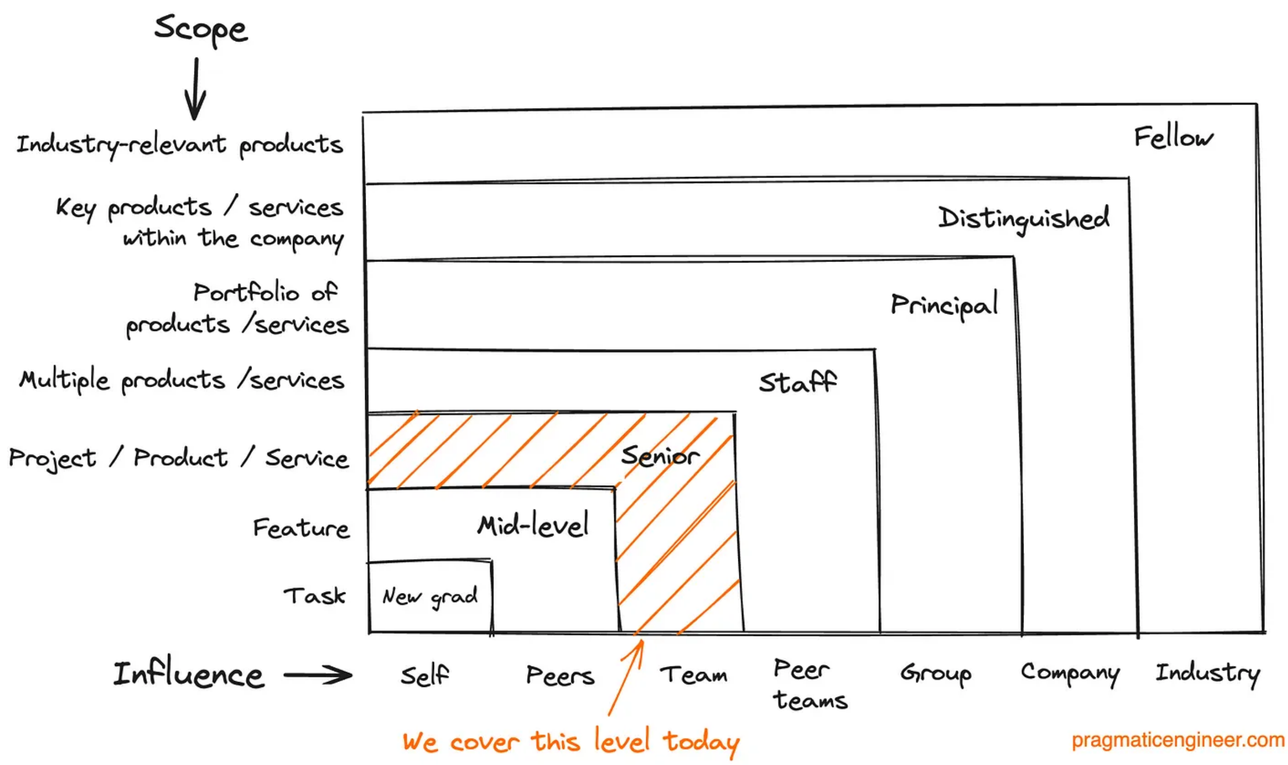
1. Two frameworks for thinking about seniority
2. Amazon’s senior definition (SDE3)
3. Meta’s IC5 level
4. Uber’s senior engineer expectations (L5)
5. Dropbox: the IC3 position
6. Shopify’s C6 level
7. Oracle: where the internal ‘principal’ position maps to ‘senior’ within the industry

In a follow-up issue, we covered What a senior software engineer means at scaleups.

1. Two frameworks for thinking about seniority

Before we jump into how larger companies define the senior engineer role, here are two frameworks on how to think about seniority.

The first is one I came up with, which looks at scope and influence. Scope refers to what type of work you take on, and how complex it is. Influence refers to people you’re expected to – or must! – influence, as you deliver work within that scope:

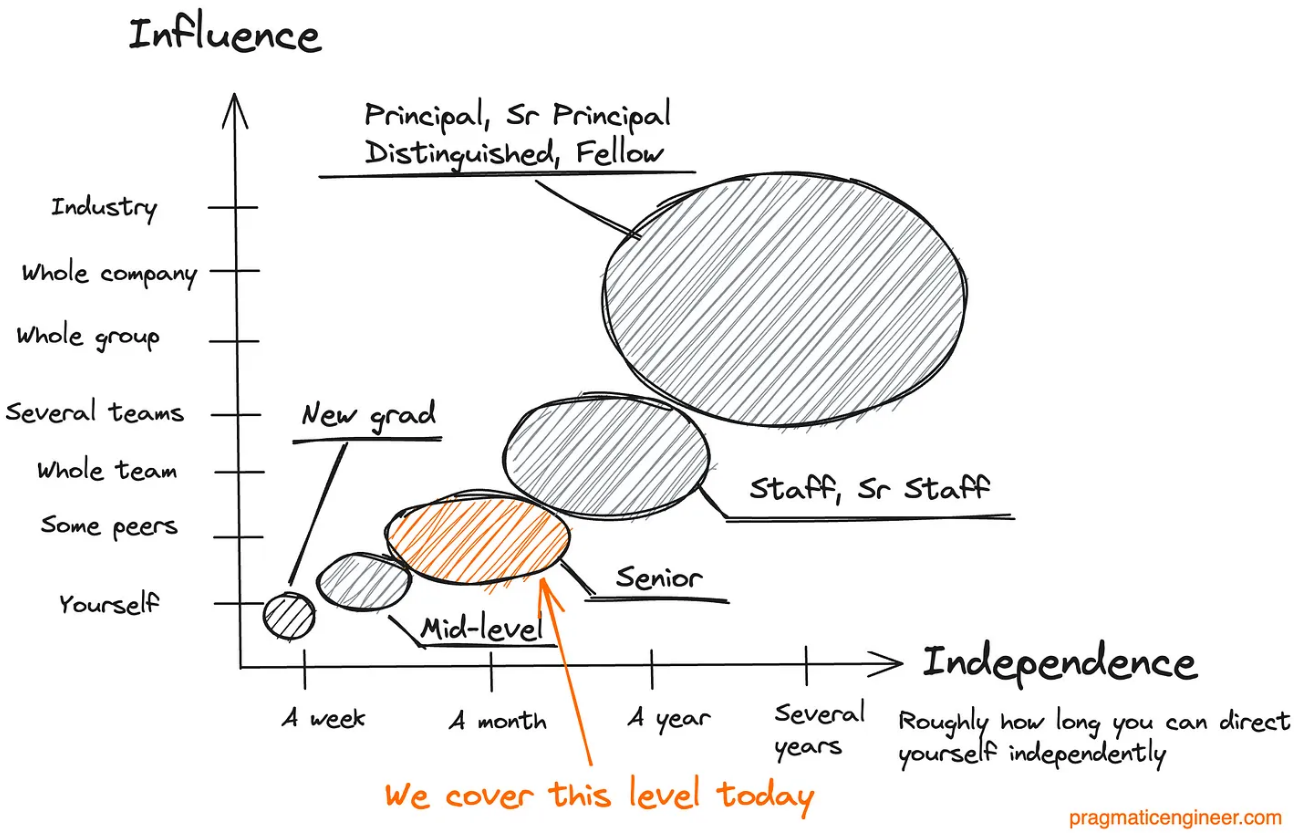


A framework for seniority using the Scope/Influence framework

The names of the levels in this framework are illustrative and titles will differ by company. The point is that the higher a position's seniority, the wider the scope of work that holders are *expected* to deliver. And to deliver this work, they *need* to be able to influence groups of a certain size.

For example, a mid-level engineer is usually expected to deliver a feature by themself, and usually don't need to influence beyond their immediate team. However, a principal engineer might be expected to make a decision on a portfolio of products their group owns, for which they need to influence the whole group. For example, making the case for moving the shared logic of several products to its own, centralized service for better maintainability.

A second framework comes from an observation shared by software engineer [Samhan Salahuddin], about the connection between influence and independence – the latter referred to as "task maturity" by Andy Grove in the book, "High Output Management." Visualizing this:



A framework for seniority using the influence/independence framework. The ideas of these axes was shared by software engineer Samhan Salahuddin

2. Amazon: SDE3

Amazon's SDE3 level maps to what the industry commonly refers to as "senior engineer." A summary of this level's typical expectations:

Area	Typical expectation
One-liner	Lead the team's software development efforts
Scope	Team and related teams
Influence	Influence team strategy and goals. Might need to influence software decisions made by other teams, to resolve blockers of their own team
Ownership	Team architecture
Time horizon	Months
Autonomy	Routinely deliver the right thing without guidance
Complexity of work	Projects where business direction is understood, but technology strategy is not
Nature of work	Spans the full lifecycle: design, implementation, testing, deployment and maintenance strategy. Develops understanding of business problems and custom cases
Engineering practices	Drives adoption of these across the team

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A summary of Amazon's SDE3 level. View this image as text here.

Amazon's SDE3 definition is closer to what is considered a "team lead" or "tech lead" role at many companies, and so I view this SDE3 role as being at the upper end of a senior engineer definition. Amazon emphasizes a few different things at this level:

- **Dealing with ambiguity.** Senior engineers are expected to tackle problems where the business problem (or customer case) *is* well defined, but the technology strategy is not. It's down to them to create well-designed, extensible, performant and secure solutions.

- **A lead role.** The SDE3 level expects leadership on projects in which this engineer is involved. However, expectations make it clear that "leading" does not mean doing all the major work. It involves mentoring colleagues, reviewing their work, and making space for others to voice opinions.
- **Writing and explaining things simply.** Amazon expects senior engineers to be skilled at communicating their technical ideas in writing. They also expect senior engineers to do this regularly. Seniors are expected to be able to explain problems and suggest solutions in simple, concise terms.
- **Autonomy**. Seniors are expected to be able to deliver the right things without guidance.
- **Maintainability**. SDE3s are expected to balance speed of execution and delivery, with building maintainable foundations for the future.

The expectation which stands out most to me, is that Amazon expects SDE3s to not be a key dependency. This is how Amazon puts it:

> "Your team is stronger because of your presence; but it does not depend on your presence to be successful."

This is a pragmatic-enough expectation which encourages mentoring and knowledge sharing. After all, if you hoard knowledge, you *will* be on the critical path for your team, so if you're absent, work can come to a halt. It's less usual for such an expectation to be spelt out this clearly, even though it's pragmatic to seek to remove the "single bus factor," wherein a team is fully dependent on an individual.

Expectations for moving into an SDE3 (senior) role

Amazon is almost alone among most tech companies in the way its internal leveling document specifies levels and their expectations, and also provides guidance on how to move to the next level. Here's the guidance Amazon offers SDE2 roles about moving to SDE3. It's not a checklist, but some expectations that could be considered:

- *Lead a complex project*. Lead a strategic team effort, starting at the design stage. This project should require multiple team members' contributions.

- *Influence the long-term roadmap*. Advocate for long-term technical investments, and build software flexible enough to evolve in future.
- *Coach and mentor*. Mentor or help others on the team.
- *Mitigate roadblocks*. Discover and mitigate unknown risks which block projects.
- *Demonstrate engineering best practices*. Work efficiently, deliver incrementally and frequently.
- *Improved writing skill*. Efficient at communicating design decisions and technical decisions in writing, and keep improving in writing for non-technical audiences.
- *Design reviews*. Lead team reviews and participate in other teams' reviews.
- *Address systemic issues*. Take the lead in identifying and solving architecture deficiencies. Reduce support costs by addressing systemic issues.

We cover more about Amazon's engineering levels in the article, Inside Amazon's engineering culture.

3. Meta: IC5

"Senior engineer" maps to the IC5 level at the social media giant. Here's what is typically expected at this level:

Area	Typical expectation
One-liner	A self-directed engineer who drives projects and meaningfully contributes to products
Scope	Project or product
Influence	Influence peers within the team. Work across teams, but rarely expected to influence at that level
Ownership	Project-level Parts of a product Service
Time horizon	A few months, to more than a year
Autonomy	Self-directed
Complexity of work	Can take on complex projects. These projects can have significant other team dependencies (e.g. many dependencies or complex dependencies)
Nature of work	Drive projects from scoping, through design, implementation and shipping. Running systems in production (oncall, consult partner teams, provide architecture feedback to downstream/upstream systems, monitoring, incident management)
Engineering practices	Contribute to improving engineering excellence across the team

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A summary of Meta's IC5 engineering level. View this image as text.

We go into more detail on engineering levels at Meta in Inside Facebook's engineering culture.

4. Uber: L5

Uber defines the senior software engineer role as its L5 level. Here's a summary of L5:

Area	Typical expectation
One-liner	Take on complex problems that span multiple related areas and teams.
Scope	Team
Influence	Across the team, and with team stakeholders (e.g. product managers, designers, operations folks)
Ownership	Project: lead certain projects and efforts across the team
Time horizon	A quarter (typically)
Autonomy	Unblocks themself when delivering projects and efforts
Complexity of work	Builds projects from scratch. Makes challenging technical decisions: e.g. buy vs build, build-from-scratch vs rewrite
Nature of work	All software lifecycle stages: idea inception, requirements gathering, design, implementation, productization, and operating it. Works with stakeholders to understand and translate complex or ambiguous customer problems.
Engineering practices	Contribute to improving engineering excellence across the team

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Uber's L5 engineering level. View this image as text here.

A few areas the company emphasizes:

- **Code quality and testing.** Writing high-quality code and tests which verify correctness and stability is heavily emphasized at the senior level.
- **Getting and giving feedback**. Gathering design feedback and code reviews, as well as providing quality feedback, is a baseline expectation.

- **Collaborate with stakeholders.** Work with product, design, operations, or the business, to understand customer problems, then translate them to systems design approaches.
- **Drive projects, end to end**. This doesn't mean from planning to shipping. It starts earlier, at the idea stage, involving stakeholders and possibly convincing them that the project makes business sense. Projects are not finished once deployed and operating the project is also a responsibility.
- **Inspire and rally others.** When leading a project, inspire other engineers, rally them to its technical goals, and create an environment where engineers have a bias for action.

Uber's engineering competencies changed during my four years there from 2016 until 2020, and have kept being refined since. Having reviewed the latest expectations, what stands out to me is how much less "individualistic" expectations for senior engineers have become over time.

For example, L5 expectations pre-2020 did not include mentions of serving as a multiplier, leading efforts to improve the efficiency of the team, building and maintaining collaborative relationships, or inspiring and rallying other engineers. It is both educational and nice to see such expectations develop.

5. Dropbox: IC3

Of the companies covered in this article, all employ more than 10,000 employees except Dropbox, which employed around 3,000 staff at the end of 2022. At the same time, Dropbox is one of the few companies which opened up its engineering framework to the public, and so I found it worthwhile to include the company in this analysis.

At Dropbox, both the IC3 and IC4 levels tend to map to what most tech companies refer to as senior software engineers. In my interpretation, IC3 is more of a senior engineer level, and IC4 is more of a team lead, or tech lead, role. In this section, we analyze the IC3 role. Here's an overview:

Area	Typical expectation
One-liner	Independently identify and deliver software solutions via a set of milestones spanning a specific product focus or a multi-component system
Scope	Projects the team owns Solves ambiguous, open-ended problems
Influence	Direct team Cross-functional partners while driving cross-team collaboration for the project
Ownership	Project
Time horizon	Quarterly
Autonomy	Unblocks themself and their team
Complexity of work	Work on systems with multiple components
Nature of work	Make timely decisions and don't cut corners. Actively keep customer needs in mind and gather input from product stakeholders
Engineering practices	Consider simplicity and maintenance when designing software. Build tools and produce technical documentation which improve developer efficiency and drive alignment within the team Adopt best practices and coding standards to foster an effective engineering culture

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The IC3 level at Dropbox. View this image as text

Dropbox open sources its engineering competencies expectations, and you can read the full IC3 software engineer definition, here. Dropbox emphasizes the following areas for senior engineers:

- **Cross-team collaboration**. IC3s are expected to be proactive and drive efforts when they work with other teams to get things done.
- **Customer focus.** Understand the goals of customers, the business and their team. Ensure their work has the greatest customer impact.
- **Taking responsibility.** IC3s own mistakes, learn from them, and take action to prevent a repeat. They embrace these learnings and share them with others.
- **Execute in the spirit of requirements**. Make informed decisions by consulting the right stakeholders and balancing details with the big picture. Execute in the spirit – not just the letter – of requirements.
- **Avoid compromising customer trust.** Don't cut corners that speed up work but reduce customer trust.
- **Remain resilient through change.** Be calm under pressure and take care of personal well-being. Navigate ambiguity by focusing on the greater purpose. One step at a time.
- **Proactive in sharing information.** Be proactive in communicating key details with stakeholders, including the team and manager. Set the right expectations upfront with your manager to balance work and other duties like mentorship.
- **Mentorship**. Active in leveling up less experienced team members. Help them with their craft and set a good example to follow.

The expectations which stand out to me cover outages and "keeping the lights on" (KLTO):

> "I am unafraid of declaring a SEV [incident] when needed. (...)
>
> I actively seek out and eliminate sources of toil on the team and help reduce the impact of KTLO and SEVs
>
> I respond with urgency to operational issues (e.g., SEVs,) owning resolution within my sphere of responsibility
>
> I proactively create and update playbooks for components I own."

SEV refers to "Severity" and "declaring a SEV" means to declare an incident. I find it a telling sign that Dropbox heavily refers to oncall practices and reducing maintenance burden ("help reduce the impact of KTLO") in its IC3 guidance. Such documents tend to indicate what companies care about most. And it's clear Dropbox cares about reliability, and also making sure oncall is a decent experience – with good playbooks to hand – which doesn't burn out people, and focuses on the impact of reading KLTO work.

6. Shopify: C6

At Shopify, the C6 level – short for 'Crafter 6' – maps most closely to what much of the industry calls a senior engineer. A summary of this role:

Area	Typical expectation
One-liner	Role model for the team who drives and delivers team projects
Scope	Team
Influence	Provide technical direction and build technical alignment within a problem space
Ownership	Project Service (which the team owns)
Time horizon	Months
Autonomy	Undertake ambiguous or poorly defined problems
Complexity of work	Ambiguous, interconnected problems
Nature of work	Define implementation strategies, estimate, drive and deliver projects. Collaborate with the product manager to define project work.
Engineering practices	Improve the team's capacity to ship great code. Balance tradeoffs and build for the long-term

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The C6 level at Shopify. View this image as text

We cover more on Shopify's leveling approach, its recent leveling split, and the concept of 'mastery' in Inside Shopify's leveling split.

7. Oracle: IC4

Oracle's IC4 – Principal Member of Technical Staff – is the role I believe is most similar to what's commonly defined as "senior."

For context, Oracle has an IC3 role – Senior Member of Technical Staff – which expects engineers to perform project tasks with minimal direction. This role tends to expect 2.5 years or more of professional experience, but there are new grads hired for this role straight from university. My feeling is that IC3 is more of a "mid-level" role across the industry, and maps to "Software Engineer 2" at many companies.

Here's how Oracle's IC4 role can be summarized:

Area	Typical expectation
One-liner	Full responsibility for simple project, or a significant, self-contained portion of a complex project
Scope	Project
Influence	Leading projects with 2-3 engineers
Ownership	Project
Time horizon	Months
Autonomy	Delivers projects independently
Complexity of work	Moderate to high conceptual complexity
Nature of work	Designs, codes, delivers and documents. Has some visibility outside their team, e.g. presenting at the group-level
Engineering practices	Writes high-quality code and documentation

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The IC4 (Principal Member of Technical Staff) level at Oracle. View this image as text

To get promoted to IC4 from IC3, support from an immediate manager is necessary. To be promoted to IC4 to IC5, managers from outside the candidate's immediate group need to support it. For IC6, peer engineers at that level also need to explicitly support it.

Takeaways

We've gathered some common expectations, and here's how the 'senior engineer' role can be summarized at many large tech companies:

Area	Typical expectation
One-liner	Independently delivers projects within scope of the team/product
Scope	Team and product
Influence	Full influence across their team Might influence beyond their team Collaborate with other teams and cross-functional roles (e.g. product managers, designers, data, and others)
Ownership	Project ownership Services the team owns
Time horizon	Months to quarters
Autonomy	Self-directed Delivers projects independently Unblocks themself and their project, as needed
Complexity of work	Take on and complete projects the team considers moderate-to-high complexity (within team scope) Can execute projects with other team dependencies, and/or multiple components
Nature of work	From planning (sometimes even before,) to shipping and operating the software/systems *The nature of work varies more across companies*
Engineering practices	Improves engineering practices, product and engineering quality across the team. Considers simplicity and maintenance when designing and building software

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A 'typical' senior engineer expectation across companies. View this image as text

Of course, all generalizations have their limits.

Roles to collaborate with vary across companies and teams. For example, at Uber, it's much more common for engineers to work with operational roles, because Uber has many of these who are core to the business, like customer support, driver support, and others. This is a reminder that engineers are exposed to different roles depending on a company's specific environment.

Senior engineers are usually expected to get *closer* to actual customers. At large companies, this usually means working with roles which directly work with customers, be they product managers, designers, operations, or others.

Also, don't forget that at larger companies it's common enough for engineers to build internal products, where the customer is another team!

The nature of the work varies per company. For example, at Meta it's much more common for engineers to get involved in a project before it's defined, at the scoping and design phase. In fact, engineers are more likely to propose projects that get built at a place like Meta, than at other bigger companies where roles are more rigid and proposals are more likely to come from product, or from above.

So there will be places where it's a given that senior engineers work on very "fuzzy" ideas, to which they bring their product sense. However, at other places the norm might be for engineers to only get involved when a product specification is complete. They work with engineering on planning, then building and shipping.

The focus on operations and maintenance is important. Several companies explicitly mention operations in their expectations for the senior engineer role, while Dropbox spells out the importance of oncall practices and engineering efforts on reliability. Being oncall is almost always a given at these companies, and senior engineers tend to be expected to improve teams' oncall and reliability cultures.

More details about healthy oncall practices.

Improving engineering practices and engineering quality is also a focus. At these companies, senior engineers are expected to get things done with good-enough quality by themselves, and to lead by example on code and documentation quality (Oracle,) or improve the engineering excellence across their team (Meta, Uber, Dropbox, Shopify.)

I expect this kind of focus – looking at the whole team – to be much less present at smaller companies and startups, where long-term, team-level maintainability is not as big a concern. However, at companies with hundreds of engineering teams, where engineers depart and onboard in a continuous stream, a healthy engineering culture within teams is essential to a well-oiled engineering organization.

Some companies separate a "tech lead" role, and others don't. At Dropbox, both the IC3 and IC4 roles can be described as senior engineer-level. The difference is that IC4 is much more of a tech lead role than the IC3 role is. This split is similar to the L5 senior engineer and L5B staff engineer levels at Uber, where L5's scope is at the project level, while L5B tends to be at team level.

However, not all companies make this separation. For example, at Amazon acting as a tech lead is part of the SDE3 role.

However, in the end it's less important what the "official" expectations are, and more important what expectations your manager – and organization – have of your role.

Don't forget that "written expectations" and actual expectations frequently diverge. While all large tech companies have a written document spelling out expectations in more detail than I've shared, keep in mind that following the paperwork is not always what makes a solid senior engineer.

A senior engineer is someone who lives up to the expectations of their team, manager and management chain in getting things done, delivering software, and being a role model.

Documentation summarizes a condensed version of the behaviors which engineering leadership *thinks* senior engineers should display. But expectations change over time and there are always exceptions. Oftentimes, written documentation is behind the latest expectations of your role.

To be sure you're performing at the level expected of you, get feedback from your manager, peers and – if possible – from mentors within the company. Every 6 or 12 months, there's more formal feedback as part of performance reviews. *We cover more on this in [Preparing for performance reviews ahead of time](#).*

I hope this overview is educational. We'll follow up on what "senior engineer" means at scaleups, startups and digital agencies in later issues.

Finally, a couple of earlier articles that are related to the topic of seniority, and you might find them relevant to read, or re-read:

- Engineering career paths at Big Tech and scaleups
- What is a Staff+ engineer?
- Engineering leadership skill set overlaps
- The seniority rollercoaster

7 Comments

lliamander Jul 5 Liked by Gergely Orosz

This mostly jives with my favorite essay on engineer career ladders: http://changelog.ca/log/2013/08/09/software_engineer_title_ladder#senior

It seems to me the most salient difference between companies on how they define "senior" is to whether or not the definition includes being a tech lead. Personally, I think it makes the most sense to keep those levels separate. "Senior Engineer" has become the career level across our industry, and while not every engineer can be expected to become a lead, it is reasonable to expect that every engineer will eventually come to the place where the can independently own tasks from beginning to end.

1 reply by Gergely Orosz

Aish Fenton Jul 5 Liked by Gergely Orosz

Do you have these tables for Staff too? Would be interesting to compare (and don't think they're in the staff+ article).

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1 reply by Gergely Orosz

5 more comments...